

Power System Protection

Assignment -1

Group -1

- a. List the basic requirements of protective relay.
- b. Define the selectivity of protective relay.
- c. What is back-up protection?
- d. Define the term – burden and over reach as related to protective relay.
- e. Define the term – pick up value of a relay.
- f. Give at least two merits of a resistance grounded system.
- g. Draw the different over current relay characteristics based on operating time.
- h. Draw the schematic diagram for the biased differential protection.
- i. What are the advantages of percentage differential relay over simple differential relay?
- j. What is the need for negative sequence relay?
- k. What are the merits of mho relay?
- l. What are the advantages of static relay over electromagnetic relay?
- m. Discuss the effect of arcing fault on the reach of simple impedance relay and reactance relay.
- n. What are the conditions to be satisfied for the operation of directional over current relay?
- o. Determine the time of operation of a relay of rating 5 amps, 2.2 sec IDMT and having relay setting of 125% with $TMS=0.6$. It is connected to a supply circuit through a CT of 400/5 ratio. The fault current is 4000A.
- p. Name the types of faults can be protected by the Buchholz Relay.
- q. How are the CTs on the primary and secondary side of a Y- Δ transformer connected for differential protection and why?
- r. What will be the PSM for the relay if the fault current is 120A, Given the plug setting is 1.2A and the C.T ratio is 50/1?
- s. Draw R-X diagram for different types of distance relay.
- t. What are the problems arising in differential protection in power transformer and how are they overcome?

Group – 2

2. Describe solid grounding and resistance grounding for earthing the neutral point.
3. Compare numeric relay with electromagnetic relay.
4. Describe with neat diagram, the different type of Electro mechanical relay.
5. Describe the construction and principle of operation of an induction type directional over current relay.

a. List the basic requirements of protective relay.

The fundamental (basic) requirements or essential qualities of a protective relay/system are:

- **Selectivity** (or Discrimination): Ability to isolate only the faulty section.
- **Reliability**: Assurance of correct operation when needed and no operation when not needed.
- **Sensitivity**: Ability to detect even small faults/minimum abnormalities.
- **Stability**: No maloperation during external faults, through-faults, or transients.
- **Speed** (Fast operation): Quick fault clearance to minimize damage and maintain stability.

b. Define the selectivity of protective relay.

Selectivity (or discrimination) is the ability of a protective relay to identify and isolate only the faulty section of the power system while leaving the healthy sections in service. It ensures that the relay distinguishes between faults within its protected zone and those outside it (or normal conditions), preventing unnecessary disconnection of healthy parts.

c. What is back-up protection?

Back-up protection serves as a secondary defense mechanism. It operates to clear a fault if the primary protection fails to do so (due to relay failure, breaker failure, etc.). Back-up relays are independent of the primary protection's failure causes and include a deliberate time delay to allow the primary protection to act first.

d. Define the term – burden and overreach as related to protective relay.

- **Burden**: The power (in VA – volt-amperes) consumed by the relay circuit at its rated current/voltage. It represents the load imposed on the current transformer (CT) or voltage transformer (VT).
- **Overreach**: A condition where the relay operates for a fault beyond its intended protected zone/reach (e.g., due to transient errors, CT saturation, or incorrect settings), causing it to trip unnecessarily.

e. Define the term – pick up value of a relay.

The pick-up value (or pick-up setting) is the minimum value of the actuating quantity (e.g., current, voltage) at or above which the relay is designed to operate/start timing. Below this value, the relay remains stable and does not operate.

f. Give at least two merits of a resistance grounded system.

Merits of a resistance-grounded (especially high-resistance grounded) system include:

1. Limits ground fault current to a low/safe value (much lower than solidly grounded systems), reducing equipment damage and arc flash hazards.

2. Provides damping to eliminate/avoid arcing grounds (persistent/intermittent faults) and improves system stability. Additional merit: Reduces inductive interference with nearby communication circuits.

g. Draw the different over current relay characteristics based on operating time.

(Note: In a Word document, insert diagrams or sketches here. Descriptions below for reference.)

Common time-current characteristics for overcurrent relays:

- **Definite Time:** Operating time is constant (independent of current magnitude above pick-up).
- **Inverse Definite Minimum Time (IDMT):** Operating time inversely proportional to current at lower values; becomes definite (minimum time) at high currents.
- **Very Inverse:** Steeper inverse curve than standard IDMT; better for selectivity in certain applications.
- **Extremely Inverse:** Very steep; often follows $I^2t = \text{constant (K)}$ characteristic; ideal for protecting equipment with thermal limits (e.g., cables, transformers).

h. Draw the schematic diagram for the biased differential protection.

(Note: In Word, insert a diagram showing two CTs on either side of protected equipment, with operating coil in the differential path and restraining/biasing coils in the through-current path.)

Biased (percentage) differential protection uses:

- Restraining coils in the circulating current path (proportional to through current).
- Operating coil in the spill/differential current path. This provides percentage bias to improve security against external faults/CT errors.

i. What are the advantages of percentage differential relay over simple differential relay?

Percentage (biased) differential relays offer:

- High sensitivity for internal faults (even low-magnitude ones).
- High security/stability during external faults or heavy through-faults (bias increases pick-up with through-current, countering CT mismatch/saturation errors). Simple differential relays may maloperate on external faults due to CT ratio errors, saturation, or tap differences.

j. What is the need for negative sequence relay?

Negative sequence relays protect rotating machines (e.g., generators, motors) from overheating due to unbalanced stator currents (e.g., from single-phase faults or open conductors). Unbalanced currents produce a negative sequence component that induces double-frequency currents in the rotor, leading to severe heating and potential damage.

k. What are the merits of mho relay?

Merits of mho (admittance) relay:

- Inherently directional (self-polarized; no separate directional element needed).
- Characteristic encloses the smallest area on the R-X diagram (circle passing through origin), making it least affected by power swings/load variations.
- Particularly suitable for long transmission lines (high load angles, power swing immunity).

l. What are the advantages of static relay over electromagnetic relay?

Advantages of static (solid-state/electronic) relays over electromagnetic (induction/disc) relays:

- Lower burden on CTs/VTs.
- Faster operating speed/response.
- No moving parts → longer life, less maintenance, higher reliability.
- Compact size.
- Higher sensitivity (easy signal amplification).
- Ability to implement complex characteristics/logic via electronic circuits.

m. Discuss the effect of arcing fault on the reach of simple impedance relay and reactance relay.

- **Impedance relay** (measures total $Z = R + jX$): Arc resistance adds to the real (R) component → apparent impedance increases → causes **underreach** (fault appears farther than actual; may fall outside the relay's circular zone).
- **Reactance relay** (measures only reactive component X): Unaffected by arc resistance (which is resistive) → reach remains accurate; no underreach.

n. What are the conditions to be satisfied for the operation of directional over current relay?

Two main conditions:

1. The fault current magnitude must exceed the relay's pick-up/setting value.
2. The fault current must flow in the forward/pre-specified direction (determined by voltage/current phase comparison for directionality).

o. Determine the time of operation of a relay of rating 5 amps, 2.2 sec IDMT and having relay setting of 125% with TMS=0.6. It is connected to a supply circuit through a CT of 400/5 ratio. The fault current is 4000A.

Verified calculation (assuming standard inverse IDMT characteristic):

1. Relay current setting = 125% of 5 A = **6.25 A**.
2. CT ratio = 400/5 = 80.
3. Secondary fault current = 4000 A / 80 = **50 A**.
4. Plug Setting Multiplier (PSM) = 50 A / 6.25 A = **8**.

5. For standard IDMT at PSM = 8 and TMS = 1, operating time is typically **~2.95–3.0 s** (common table value ≈ 3.0 s).
6. Actual operating time = $3.0 \text{ s} \times \text{TMS } 0.6 = \mathbf{1.8 \text{ seconds}}$.

p. Name the types of faults can be protected by the Buchholz Relay.

Buchholz relay protects oil-immersed power transformers against incipient/internal faults, including:

- Minor arcing in oil.
- Insulation deterioration/decomposition.
- Hot spots in core/windings.
- Major internal faults causing rapid gas generation (for instantaneous trip).

q. How are the CTs on the primary and secondary side of a Y-Δ transformer connected for differential protection and why?

- CTs on the star (Y) side are connected in **delta (Δ)**.
- CTs on the delta (Δ) side are connected in **star (Y)**. Reason: This compensates for the inherent 30° phase shift in Y-Δ transformers and blocks zero-sequence currents from causing maloperation during external/through faults.

r. What will be the PSM for the relay if the fault current is 120A, Given the plug setting is 1.2A and the C.T ratio is 50/1?

1. Secondary fault current = $120 \text{ A} / 50 = \mathbf{2.4 \text{ A}}$.
2. PSM = Secondary current / Plug setting = $2.4 \text{ A} / 1.2 \text{ A} = \mathbf{2}$.

s. Draw R-X diagram for different types of distance relay.

(Note: Insert diagrams in Word.)

- **Impedance relay**: Circle centered at origin.
- **Reactance relay**: Straight line parallel to R-axis (constant X).
- **Mho relay**: Circle passing through origin (diameter along line angle).
- **Angle impedance relay**: Straight line at an angle to axes.

t. What are the problems arising in differential protection in power transformer and how are they overcome?

Main problems and solutions:

1. **Magnetizing inrush current** (high during energization): Overcome by harmonic restraint (2nd harmonic blocking/filtering).
2. **Tap changing** (causes ratio mismatch/unbalance): Overcome using percentage (biased) differential relays with slope/ bias setting.
3. **Phase shift in Y-Δ connections**: Overcome by connecting CT secondaries in Δ-Y configuration to compensate 30° shift and block zero-sequence currents.

2. Describe solid grounding and resistance grounding for earthing the neutral point.

Solid Grounding (also known as Effectively Grounded or Solidly Grounded)

- **Definition:** The neutral point of a generator, transformer, or system is connected directly to earth (ground) with no intentional impedance (resistance or reactance) in the path. A system is classified as **effectively grounded** per IEEE Std 142 / IEEE C62.92.1 when the ratios satisfy:
 - Zero-sequence reactance to positive-sequence reactance: $X_0/X_1 \leq 3$ (and positive).
 - Zero-sequence resistance to positive-sequence reactance: $R_0/X_1 \leq 1$ (and positive).
- **Key Characteristics:**
 - Coefficient of grounding (COG) $\leq 80\%$ (highest healthy phase-to-ground voltage during L-G fault $\leq 80\%$ of line-to-line voltage).
 - High ground fault current (often comparable to or exceeding three-phase symmetrical fault current).
 - Eliminates arcing grounds by fully neutralizing capacitive currents with inductive fault current.
 - Limits transient overvoltages effectively.
- **Advantages:**
 - Easy detection and fast clearance of ground faults via high current levels.
 - Good control of transient overvoltages.
 - Suitable for systems with phase-to-neutral loads.
- **Disadvantages:**
 - High fault currents cause significant equipment damage, arc flash hazards, and mechanical stress.
 - Not ideal for continuity-critical systems.
- **Applications:** Common in low-voltage systems (≤ 600 V) and high-voltage transmission/distribution (> 33 kV or effectively grounded medium-voltage systems).

Resistance Grounding

- **Definition:** The neutral is connected to earth through a deliberate resistor (R), which limits ground fault current.
- **Key Characteristics:**
 - Fault current is controlled by resistor value (typically low-resistance: 200–400 A; high-resistance: 5–10 A or near capacitive current level).
 - Neutral displacement occurs during L-G faults; healthy phase voltages can rise to line-to-line value if not tuned properly.
 - Can be tuned to make lagging fault current neutralize capacitive current, suppressing arcing grounds.
- **Advantages:**
 - Significantly reduces ground fault current compared to solid grounding → less damage, lower arc flash risk, reduced mechanical/thermal stress.
 - Damping effect improves stability and eliminates persistent arcing grounds.
 - Minimizes inductive interference with nearby communication circuits.
- **Disadvantages:**
 - Requires careful resistor sizing for relay operation and fault detection.

- May need additional measures for full overvoltage control in some cases.
- **Applications:** Predominantly in medium-voltage industrial/commercial systems (3.3 kV to 33 kV); low-resistance for selective coordination, high-resistance where minimal fault current is desired.

3. Compare numeric relay with electromagnetic relay.

(Verified and expanded comparison table based on standard sources; insert as a formatted table in Word.)

Feature	Electromagnetic Relay	Numerical (Digital) Relay
Size & Construction	Larger, bulkier; mechanical moving parts	Compact, solid-state/microprocessor-based
Characteristics	Fixed (hardware-based)	Programmable/selectable; multiple curves/functions
Flexibility	Low (limited to one or few functions)	High (multi-function, configurable via software)
Communication	None or limited	Extensive (IEC 61850, Modbus, SCADA integration)
Self-Supervision	None	Continuous self-diagnostics and monitoring
Accuracy	$\pm 5\%$ or more	High ($\pm 1-2\%$)
Speed of Operation	Relatively slower	Very fast (milliseconds)
Burden on CTs/VTs	High	Very low
Maintenance	Frequent (contacts, pivots, calibration)	Minimal/maintenance-free
Memory/Features	None	Event recording, fault locator, oscillography
Spares & Cost	Many specific spares needed	Universal modules; lower long-term cost
Reliability	Lower (mechanical wear)	Higher (no moving parts)

Numerical relays have largely replaced electromagnetic types in modern systems due to superior performance, multifunctionality, and integration capabilities.

4. Describe with neat diagram, the different types of Electro mechanical relay.

Electromechanical relays use mechanical movement produced by electromagnetic forces. Principal types:

- **Attracted Armature Relays** (operate on attraction force; AC/DC compatible):
 - **Hinged Armature Type:** Coil flux attracts hinged iron armature to close contacts (simple, fast for instantaneous applications).
 - **Plunger Type:** Solenoid pulls plunger inward against spring to operate contacts.

(Diagram note: In Word, insert sketch showing electromagnet coil, hinged/plunger armature, contacts, and spring return.)

- **Induction Relays** (AC only; operate on induction torque like induction motor):
 - **Induction Disc Type:** Aluminum disc rotates between electromagnet poles (shaded-pole or wattmeter structure creates phase-shifted fluxes for torque). Used for time-overcurrent.
 - **Induction Cup Type:** Hollow aluminum cup rotates over stationary core; high torque/inertia ratio → fast, accurate for directional/distance relays.

(Diagram note: Sketch shaded-pole disc with upper/lower magnets, shading rings; cup type with cup over core poles.)

- **Thermal Relays:** Use heating effect on bimetallic strip (bends to operate contacts). Primarily for motor overload protection (time-delay based on Pt).

(Diagram note: Show bimetal strip, heater coil, trip contacts, and reset mechanism.)

5. Describe the construction and principle of operation of an induction type directional overcurrent relay.

Construction The relay combines two units in one case:

- **Directional Unit:** Induction cup (or wattmeter) type; energized by voltage (polarizing) via VT and current via CT.
- **Overcurrent Unit:** Induction disc (shaded-pole or watt-hour meter) type; current-operated.
- **Control Circuit:** Directional unit contacts connected in series with overcurrent unit secondary winding (keeps it open until direction is correct).

(Diagram note: In Word, insert schematic showing: CT → overcurrent disc electromagnet; VT → directional cup electromagnet; directional contacts in series with overcurrent secondary circuit; final trip contacts from disc.)

Principle of Operation

- **Directional Sensing:** Directional unit produces torque based on phase angle between voltage (polarizing) and current. Torque is positive (contacts close) only for forward power flow (fault in protected direction, phase angle within $\sim \pm 90^\circ$ of characteristic angle). Reverse faults produce no/negative torque (contacts open).
- **Overcurrent Sensing:** When directional contacts close (forward fault), overcurrent unit's secondary circuit completes. If fault current > pick-up setting, disc rotates due to induction torque (time-delayed per characteristic).
- **Tripping:** Disc rotation closes trip contacts after time delay, operating breaker.
- **Selectivity:** No operation for: (a) reverse faults (directional unit blocks), or (b) forward faults below overcurrent pick-up.

This ensures directional discrimination with inverse-time overcurrent protection.